

SANDEEP KUMAR JHA

Third Year Undergraduate
Department of Economic Science

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Academic Qualifications

Year	Degree/Certificate	Institute	Performance
2023 - Present	B.S. - Economics	Indian Institute of Technology Kanpur	7.0/10
2022	CBSE (XII)	Progressive Central High School Patsa	88.4%
2020	CBSE (X)	Progressive Central High School Patsa	94.2%

Academic Achievements

- Secured **All India Rank 6171** in **JEE Advanced** among the 1.8 Lakh shortlisted candidates from across the country 2023

Professional Experience

AI Tools & Full Stack Developer Intern | DigitX Apr'25 - May'25

- Built an intelligent image generation tool using **OpenAI DALL-E 3**, **Piapi**, and **Flux API** to automate dynamic AI-driven image creation
- Built a stop button using **AbortSignal**, allowing users to interrupt **AI response generation** and giving them better control and flexibility
- Added **prompt editing** and retry options so users can change previous prompts and generate updated AI responses without starting over
- Built AI-powered **Chart** and **Diagram generation** using **Chart.js** for interactive charts and **Mermaid** for flowchart and sequence diagram
- Built AI-powered **Video generation tools** using **Kling** and **PixVerse APIs** to create short, high-quality videos from user prompts

Founding Developer | NextRoundAI Jun'25-present

- Built an **AI Mock interview** platform using **Whisper** for speech-to-text and OpenAI GPT to generate interview questions based on PYQs
- Integrated a **Virtual avatar** experience using **Ready Player Me** and **Google TTS** to deliver synchronized visual and audio responses
- Developed a **Video analysis** module with **OpenCV** to evaluate user expressions and confidence, enhancing personalized feedback quality
- Created a custom **Python evaluation engine** to generate detailed **performance scores**, helping users identify improvement areas

Mobile App Developer | Freelancer | Chinar Dairy May'25 - Jun'25

- Developed three **Cross-platform mobile apps** for customers, admins, and delivery personnel using **Flutter** and **Firebase**
- Built the customer app with features like **product ordering**, **flexible subscription services**, **secure payments** using **Razorpay Payment Gateway**, a referral code system, order history tracking, and real-time **push notifications** via **Firebase Cloud Messaging**
- Implemented **smart delivery tracking** using **Geolocator** to calculate accurate distances between locations, and displayed **live delivery routes** and status markers using customizable map tiles, with smooth **navigation integration** via **Google Maps** and other apps
- Created the admin app for **Management** of product prices, stock availability, and delivery staff tasks, streamlining day-to-day operations

Full Stack Developer | Freelancer | A2Z Marketing Dec'24-Jan'25

- Developed and deployed the **A2Z Marketing website** using **React.js** on **Vercel**, with a responsive UI and optimized load performance
- Built reusable UI components like **Navbar**, **Image slider**, **Banners**, and a **Contact footer** with **integrated map** and info sections

Key Projects and Workshops

Digitizing Entry With Flutter | Project Mentor| ACA May'25-Jul'25

- Developed a feature-rich **Customer App** using **Flutter** and **Firebase** with features like **Ticket booking**, **Wallet system**, **Referral codes**, **Loyalty points**, and secure online payments using **razorpay_flutter** and **qr_flutter** for fast and reliable QR code-based ticketing
- Built Admin app for park staff to **Manage ticket pricing**, time slots, offers, and perform **QR code scanning** using **mobile-scanner**
- Integrated a range of **Firebase services**, including **firebase_auth** for secure **OTP-based login**, **cloud_firestore** for storing and retrieving real-time data, **firebase_messaging** for handling push notifications, and **flutter_local_notifications** for displaying timely in-app alerts

Bayesian Language Model | Course Project Jan'25 - Apr'25

Prof. Himanshu Yadav | CGS698C - Bayesian Models & Data Analysis

- Developed a Bayesian language model for text prediction using **R**, **Stan**, and **Python**; addressed data sparsity via smoothing techniques
- Applied **MCMC**, **HMC**, and **hierarchical modeling** to improve text prediction accuracy through robust **Bayesian inference** methods
- Optimized model performance with **hyperparameter tuning**, evaluated using key metrics like **perplexity** and **prediction accuracy**

Codeforces Recent Action | Programming Club, IIT Kanpur Feb'24

- Built a **web server** using **Go** and used **MongoDB** for structured storage and fast querying of Codeforces blog updates
- Integrated** concurrent data fetching using **goroutines** and background workers to poll the Codeforces API without blocking main server
- Dockerized** the entire application for easy **deployment**, automatic database sync, and efficient resource usage across environments

Hide Out Lounge | Self Project Dec'25

- Built two Android apps using **Flutter** for **cross-platform**, integrating **Firebase** for backend, user authentication, and admin management
- Designed **scalable Firestore** data models to support dynamic menu changes, live order tracking, and fast, efficient data retrieval
- Integrated **Firebase Auth** with **OTP authentication** to handle secure user login and provide order status updates via push notifications
- Deployed on the **Google Play Store**, reaching over **500+ downloads** and showing strong customer engagement and adoption

Bayesian Credit Risk Analyzer | Course Project Jan'25 - Apr'25

Prof. Himanshu Yadav | CGS698C - Bayesian Models & Data Analysis

- Built a Bayesian credit risk prediction model using **R**, **Stan**, **brms**, and **Python**, applying **MCMC**, **HMC**, and **hierarchical modeling**
- Predicted loan default using **Bayesian logistic regression**, improving accuracy via **cross-validation** and **borrower feature analysis**
- Tuned model hyperparameters** and thoroughly evaluated model performance using accuracy metrics for robust loan default prediction

- **Developing a cross-platform mail client** using **Flutter and Dart**, optimized for **Android, iOS, Windows, macOS, and Linux**
- Integrated **email protocols** like **SMTP, IMAP, and MBOX** using the **enough_mail** package to enable smooth email management
- **Implemented** reliable **local email storage** with **ObjectBox** for offline access, quick search functionality, and efficient email indexing
- Empowering 9K+ IITK campus janta with a modern, productive email solution designed for multi-device access and efficient communication

TECHNICAL SKILLS

- **Languages:** C/C++, Go lang, Python, JavaScript, TypeScript, Dart, HTML, CSS, R, LaTeX,
- **Software/Technologies:** Git, Postman, Firebase, PostgreSQL, MongoDB, Docker, REST APIs, RStudio, Visual Studio, Cursor, Vercel
- **Framework/Libraries:** React.js, Next.js, Go-gin, FastAPI, Flutter, Node.js ,razorpay_flutter ,flutter_local_notifications, qr_flutter

RELEVANT COURSES

Fundamentals Of Computing	Bayesian Models & Data Analysis	Quantum Computing
Applied Probability and Statistics	Linear Algebra And ODE	Introduction to Electronics

Positions of Responsibility

Web Manager | Inter IIT Sports Meet

Sep'24-Dec'24

- **Led development** of the official **IISM'24 website**, working with the Event Head and guiding a team to build a platform for **4000+** users
- **Directed real-time modules** for **live scores**, dynamic brackets, schedules, and announcements, ensuring users received instant updates
- **Implemented a real-time backend system** to **manage** and update **scores, brackets, and notifications** seamlessly during matches
- **Built a custom meal tracking system** to efficiently **manage meal access** for **4000+ registered users** during the event

Web Executive | Student Placement Office (SPO)

Jun'24-Apr'25

- Maintained the **RAS portal** and **SPO website** using **Next.js** and **GoLang**, serving 3K+ active campus users during the placement season
- Built **document and resume verification** features on the **RAS portal** to streamline the placement and internship application process

Extra-Curricular Activities

- **Freelancing on Upwork** as a Mobile and Web Developer; successfully completed **international client** projects and earned in **USD**
- Participant in **Takneek '23** and **Takneek '24**, contributed to solving **technical problem statements** during the inter-hall competition
- Dance **Volunteer at Antaragni '23**; handled backend operations for the dance event, ensuring smooth coordination and execution